# Chetan Ganeshbhai Sakre

cs25mtech14026@iith.ac.in | chetansakre123@gmail.com | +91 7984559979 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.88 |
| Gyanjyot Vidhyalaya | Class XII (GSEB) | 2020 | 82.15% |
| Gyanjyot Vidhyalaya | Class X (GSEB) | 2018 | 89.67% |

## WORK EXPERIENCE

**Tata Consultancy Services (TCS) - Digital**

*SOC Analyst -- Department of Post (DOP), CSI Operations Security Project*

- **Analyzed** large-scale enterprise security logs using structured anomaly-detection approaches to identify **suspicious activities and security incidents**.
- **Automated** investigation workflows using scripting, streamlining **security alert analysis and incident response**.
- **Investigated** security events using **RSA NetWitness and Unix/Linux tools**, collaborating with engineering teams to **identify root causes and resolve incidents**.

## PROJECTS

**Scalable Multi-Agent LLM System for Automated Feature Engineering | Prof. Sobhan Babu**

*Python, PyTorch, XGBoost, Scikit-Learn, Asyncio, LLM APIs*

- **Engineered** a 6-agent LLM framework for automated feature generation and validation on tabular data.
- **Reduced training passes by 60× and achieved 10.6× speedup** using stratified proxy evaluation and vectorized permutation scoring, scaling to **101,766 rows**.
- **Built fault-tolerant guardrails** with KS tests for covariate drift and an **asyncio 180s watchdog**, improving ROC-AUC across **4 benchmark domains**.

**WhatsApp-Scale Distributed Real-Time Messaging Architecture | Github | Live Demo**

*Tech Stack: Python, FastAPI, WebSockets, Apache Kafka, Redis, PostgreSQL, Docker, Render*

- **Architected** a distributed messaging system targeting **10M CCU and 70K peak msgs/sec**, using WebSockets and Redis Pub/Sub for horizontally scalable real-time communication.
- **Engineered** Kafka-based asynchronous pipelines with **128 conversation_id-keyed partitions**, at-least-once delivery, ordering guarantees, and DLQ-based fault tolerance.
- **Implemented Sent Delivered Read** lifecycle, offline message delivery, read receipts, and media handling with **>80% payload reduction** through client-side image compression.
- **Hardened** the platform with **Signal Protocol E2EE**, quorum-based fencing, Docker deployment, and **309 tests**.

**FraudShield: Real-Time Financial Fraud Risk & Decisioning Engine | Github**

*Tech Stack: Python, FastAPI,XGBoost, Scikit-learn, Streamlit, Pytest, Pydantic, Docker*

- **Architected** a fraud risk engine on **284K transactions** with **578:1 class imbalance**, achieving **0.817 PR-AUC** and **87.8% fraud recall**.
- **Optimized** cost-sensitive decision thresholds using a **$250 FN / $15 FP / $2 TP** cost matrix, reducing expected fraud losses by **33.4%** and saving **$2,038**.
- **Enforced** zero-leakage evaluation with a **3-way train/validation/test split** and validation-frozen thresholds; delivered **8.6ms p50 E2E latency** with FastAPI and TreeSHAP.

**EduVerse -- Full-Stack Learning Management System | Github**

*Python, Django, MySQL, HTML, CSS, JavaScript, Bootstrap, OpenCV, Docker.*

- **Architected** DRF APIs for course catalogs, quizzes, reviews, and enrollments with pagination, search, filtering, and validation.

- **Migrated** the database from SQLite to **MySQL 8.0** and optimized ORM queries using **select_related()** and **prefetch_related()**, eliminating N+1 bottlenecks.
- **Integrated** Razorpay with **HMAC-SHA256**, automated certificates via **ReportLab/OpenCV**, and Docker deployment with **49 automated tests**.

**Autonomous Enterprise Vendor Risk & Security Auditor** | **GitHub**

*Python, LangGraph, Hybrid RAG, MCP, Qwen 122B, vLLM*

- **Automated** audits of **100+ page SOC 2 reports** with a LangGraph agentic workflow, reducing analysis time from **3-5 days to <15 sec**.
- **Engineered** a hybrid RAG pipeline using **dense retrieval, BM25, and RRF** for evidence-grounded compliance analysis.
- **Integrated MCP and Human-in-the-Loop** tooling with **Qwen 122B on vLLM** for enterprise workflow automation.

## SKILLS

**Programming Languages:** C, C++, Python, Java, SQL

**Web Technologies:** HTML, CSS, JavaScript, Django, Django REST Framework

**Machine Learning & Deep Learning:** Scikit-learn, PyTorch, XGBoost, TensorFlow, Transformers, RNN, LSTM

**Networking & Security Tools:** Wireshark, RSA NetWitness

**Tools & Platforms:** Git, Docker, Linux, VS Code

## SCHOLASTIC ACHIEVEMENTS

- **Ranked 2nd in Gujarat** in the B.E. CSE 8th-semester examination conducted by Gujarat Technological University (GTU).
- **Solved 850+ Data Structures & Algorithms** problems across LeetCode and GeeksforGeeks.
- **MYSY Scholar** during B.E., awarded the Mukhyamantri Yuva Swavalamban Yojana scholarship.

## RELEVANT COURSES

| | | | |
|---|---|---|---|
| Deep Learning | Visual Computing | Advanced Computer Networks | Foundations of Machine Learning |
| Fraud Analytics | Operating Systems | Object-Oriented Programming | Computer Organization & Architecture |
| Network Security | Artificial Intelligence | Database Management Systems | Advanced Algorithms and Data Structures |

## EXTRACURRICULAR

Active participant in the Clean India Drive at IIT Hyderabad.